EEC 201 CA 1

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April 3, 2026

m4.1

Problem m4.1

$$h[n] = Ar^n \cos(\omega_0 n + \phi)\, u[n].$$

(a)

I first determined the transfer function corresponding to the impulse response in (1). Using standard results of z-transform for a damped cosine multiplied by $u[n]$, the system function was written as

$$H(z) = \frac{A\cos\phi - Ar\cos(\phi - \omega_0)\, z^{-1}}{1 - 2r\cos(\omega_0)\, z^{-1} + r^2 z^{-2}}.$$

I implemented this result in a MATLAB function `coef(A,r,omega0,phi)` that returned the numerator and denominator coefficient vectors b and a suitable for use with `filter`.

(b)

I verified that `coef()` works correctly by using it in conjunction with the MATLAB function `filter()` to compute the impulse response of the system. The impulse response was computed by applying `filter(b,a,x)` with $x[n] = \delta[n]$ over the interval $0 \le n \le 50$.

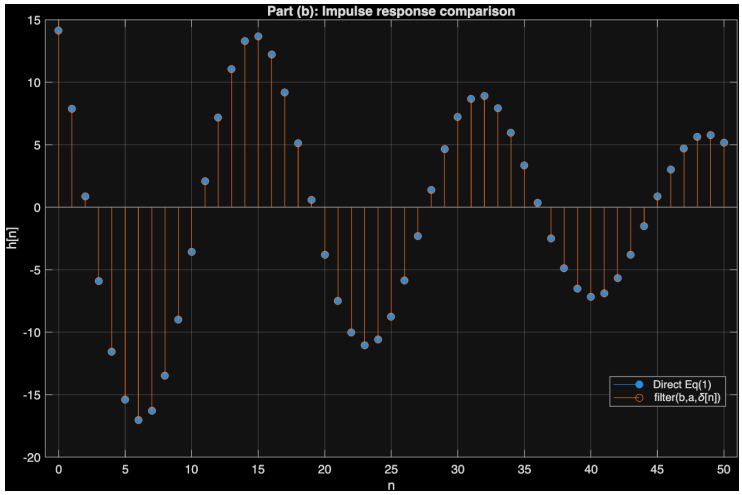
For $A = 20$, $r = 0.975$, $\omega_0 = 2\pi/17$, and $\phi = \pi/4$, the function `coef()` returned the numerator and denominator coefficients

$$b = [14.1421 \quad -17.8385], \qquad a = [1 \quad -1.8183 \;\; 0.9506].$$

I also got the impulse response by direct evaluation of Equation (1),

$$h[n] = Ar^n \cos(\omega_0 n + \phi), \qquad 0 \le n \le 50.$$

The two impulse responses were plotted on the same graph. I also used the function stem over implot!



(c)

I used `residuez` to obtain a partial fraction expansion of the system function $H(z)$ into first-order terms. Using the same system coefficients as in part (b), MATLAB returned

$$[R, P, K] = \texttt{residuez}(b, a),$$

with

$$R_1 = 7.0711 + j\,7.0711, \quad R_2 = 7.0711 - j\,7.0711,$$
$$P_1 = 0.9092 + j\,0.3522, \quad P_2 = 0.9092 - j\,0.3522,$$

and no direct term ($K = [\,]$).
Therefore, the system function can be written as a sum of first-order terms as

$$H(z) = \frac{R_1}{1 - P_1 z^{-1}} + \frac{R_2}{1 - P_2 z^{-1}}.$$

Since the system is causal, the corresponding impulse response is obtained from the power-series expansion as

$$h[n] = R_1 P_1^n u[n] + R_2 P_2^n u[n].$$

Because the residues and poles occur in complex-conjugate pairs, the impulse response is real-valued for all n and may equivalently be written as

$$h[n] = 2\,\Re\{R_1 P_1^n\}\,u[n].$$

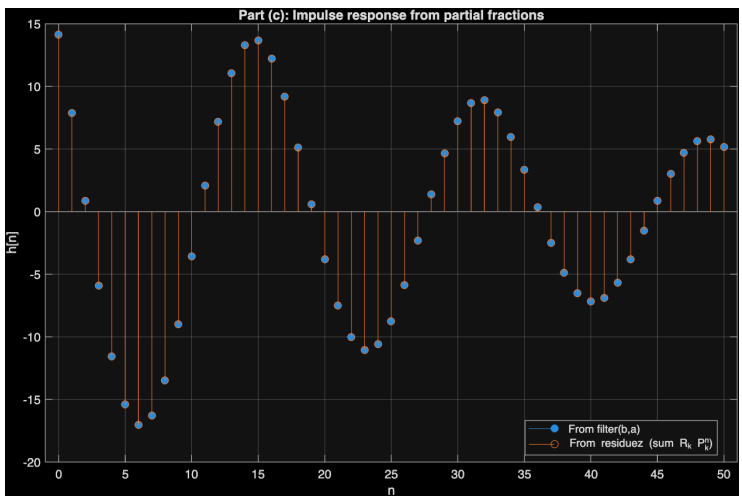
Using this expression, I computed and plotted the impulse response over the interval $0 \le n \le 50$, and compared it with the impulse response obtained

directly using `filter(b,a)` in part (b). The two impulse responses agreed to numerical precision.
The MATLAB statements used to perform this computation were

```
[R,P,K] = residuez(b,a);
```

```
n = 0:50;
```

$$\texttt{h[n]} = \sum_k R_k P_k^n, \quad n \geq 0.$$



(d)

I next used `freqz` to compute the frequency response for fixed parameters $A = 20$, $\omega_0 = \frac{2\pi}{17}$, and $\phi = 0$, using $N = 501$ frequency samples. I evaluated the response for pole radii $r = 0.9$, 0.95, 0.975, and 0.99.

From the magnitude responses, I got the resonance peak location, peak magnitude, and the corresponding 3 dB bandwidth $\Delta\omega$:

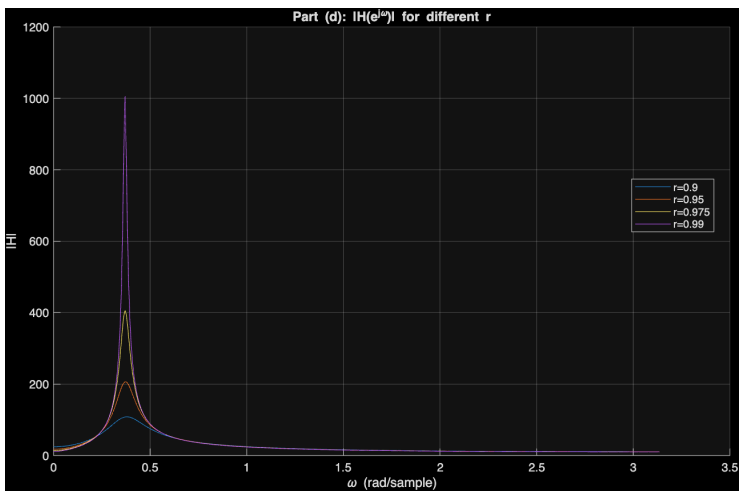
r	$1-r$	Peak location (rad)	Peak $\lvert H\rvert$	3 dB BW $\Delta\omega$
0.900	0.100	0.3825	108.18	0.2195
0.950	0.050	0.3700	206.46	0.1129
0.975	0.025	0.3700	405.83	0.0564
0.990	0.010	0.3700	1005.07	0.0251

As the pole radius r increased toward unity, the resonance peak became significantly taller and narrower. In particular, the 3 dB bandwidth $\Delta\omega$ decreased approximately in proportion to $1-r$. This behavior is consistent with the theoretical relationship for second-order resonant systems, where the bandwidth is proportional to the pole damping, which is controlled by $1-r$. Thus, poles

closer to the unit circle produce higher-Q, more selective resonances.

I also found the relationship between the resonance frequency and ω_0. For values of r close to unity, the resonance peak occurred very near ω_0. However, the peak is not guaranteed to occur exactly at ω_0 for smaller values of r. When the pole radius is reduced, the resonance broadens and the peak location can shift slightly due to the influence of the numerator and the reduced sharpness of the pole-induced resonance.

As $r \to 1$, the pole dominates the frequency response and the resonance peak converges to ω_0. Therefore, while the peak is approximately at ω_0 for large r, it is not exactly located at ω_0 in general.



(e)

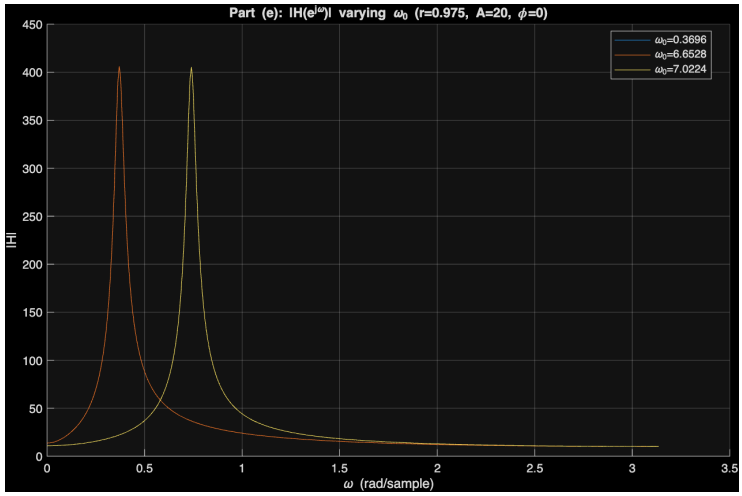
I fixed the parameters $r = 0.975$, $A = 20$, and $\phi = 0$, and varied the resonant frequency ω_0. I then plotted the frequency response for several values of ω_0, including $\omega_0 = \frac{2\pi}{17}$, $\omega_0 = \frac{2\pi \cdot 18}{17}$, and $\omega_0 = \frac{2\pi \cdot 19}{17}$.
I noticed that the frequency responses for $\omega_0 = \frac{2\pi}{17}$ and $\omega_0 = \frac{2\pi \cdot 18}{17}$ were identical. This is because the system is determined by pole locations of the form $z = re^{j\omega_0}$, and complex exponentials are 2π-periodic:

$$e^{j(\omega_0 + 2\pi)} = e^{j\omega_0}.$$

Since $\frac{2\pi \cdot 18}{17} = 2\pi + \frac{2\pi}{17}$, these two values of ω_0 correspond to the same pole angle in the z-plane, and therefore produce the same transfer function and frequency response.
For $\omega_0 = \frac{2\pi \cdot 19}{17}$, we can write

$$\frac{2\pi \cdot 19}{17} = 2\pi + \frac{4\pi}{17},$$

which shows that this case is equivalent to a resonance at $\omega_0 = \frac{4\pi}{17}$ after wrapping modulo 2π. As a such, the resonance peak appears near $\frac{4\pi}{17}$ in the plotted frequency response. This frequency lies within $[0, \pi]$.



Code: `coef.m`

```
function [b, a] = coef(A, r, omega0, phi)
a = [1, -2*r*cos(omega0), r^2];
b = [A*cos(phi), -A*r*cos(phi-omega0)];
end
```

Code: `m41.m`

```
A      = 20;
r      = 0.975;
omega0 = 2*pi/17;
phi    = pi/4;

N = 50;
n = 0:N;

% (a)
[b,a] = coef(A,r,omega0,phi);

% (b)
disp('(b) filter() coefficients ---');
disp('b ='); disp(b);
disp('a ='); disp(a);
```

```
x = [1 zeros(1,N)];
h_filt = filter(b,a,x);
h_dir  = A*(r.^n).*cos(omega0*n + phi);

figure;
stem(n,h_dir,'filled'); hold on;
stem(n,h_filt,'o');
grid on;
xlabel('n'); ylabel('h[n]');
legend('Direct Eq(1)','filter(b,a,\delta[n])','Location','best');
title('Part (b): Impulse response comparison');

fprintf('max|h_{dir}-h_{filt}| = %.3e\n', max(abs(h_dir - h_filt)));

function [b, a] = coef(A, r, omega0, phi)
a = [1, -2*r*cos(omega0), r^2];
b = [A*cos(phi), -A*r*cos(phi-omega0)];
end


% (c)
[R,P,Kd] = residuez(b,a);

disp('(c) residuez results');
disp('R ='); disp(R);
disp('P ='); disp(P);
disp('K ='); disp(Kd);

h_pf = zeros(size(n));
for ii = 1:length(R)
    h_pf = h_pf + R(ii) * (P(ii).^n);
end

if ~isempty(Kd)
    for m = 1:length(Kd)
        h_pf = h_pf + Kd(m) * (n == (m-1));
    end
end

h_pf = real(h_pf);

figure;
stem(n, h_filt, 'filled'); hold on;
stem(n, h_pf, 'o');
grid on;
xlabel('n'); ylabel('h[n]');
```

```
legend('From filter(b,a)', 'From residuez', 'Location', 'best');
title('(c) Impulse response from partial fractions');

% part (d)
A_d   = 20;
phi_d = 0;
omega0_d = 2*pi/17;
Nfreq = 501;
r_list = [0.9, 0.95, 0.975, 0.99];

fprintf('\n (d): Table values \n');
fprintf('%8s %10s %16s %14s %18s\n','r','1-r','Peak w (rad)','Peak \\
|H|','3dB BW \omega (rad)');

figure; hold on; grid on;
title('Part (d): |H(e^{j\omega})| for different r');
xlabel('\omega (rad/sample)'); ylabel('|H|');

for i = 1:numel(r_list)
    ri = r_list(i);
    [bi,ai] = coef(A_d,ri,omega0_d,phi_d);
    [H,w] = freqz(bi,ai,Nfreq);
    mag = abs(H);

    [pk, idx] = max(mag);
    wpk = w(idx);
    thr = pk/sqrt(2);

    iL = idx;
    while iL > 1 && mag(iL) >= thr
        iL = iL - 1;
    end

    iR = idx;
    while iR < numel(mag) && mag(iR) >= thr
        iR = iR + 1;
    end

    iL = max(iL,1);
    iR = min(iR,numel(mag));
    bw = w(iR) - w(iL);

    fprintf('%8.3f %10.3f %16.6f %14.6f %18.6f\n', ...
        ri, 1-ri, wpk, pk, bw);

    plot(w,mag);
```

```
end
legend(strcat("r=",string(r_list)),'Location','best');

% part (e)
A_e   = 20;
r_e   = 0.975;
phi_e = 0;
omega_list = [2*pi/17, 2*pi*(18)/17, 2*pi*(19)/17];

figure; hold on; grid on;
title('Part (e): |H(e^{j\omega})| varying \omega_0 (r=0.975, A=20, \phi=0)');
xlabel('\omega (rad/sample)'); ylabel('|H|');

wpk_list = zeros(size(omega_list));
w0eq_list = zeros(size(omega_list));

for k = 1:numel(omega_list)
    om = omega_list(k);
    [be,ae] = coef(A_e,r_e,om,phi_e);
    [He,w] = freqz(be,ae,Nfreq);
    mag = abs(He);

    plot(w,mag);

    [~, idx] = max(mag);
    wpk_list(k) = w(idx);

    w0eq = mod(om + pi, 2*pi) - pi;   % map to (-pi, pi]
    w0eq = abs(w0eq);                % freqz shows [0, pi]
    w0eq_list(k) = w0eq;
end

legend(strcat("\omega_0=",string(omega_list)),'Location','best');

fprintf('\n (e): Accounting for \omega_0 cases \n');
fprintf('%18s %18s %18s\n','omega0 (rad)','equiv in [0,pi]','peak from freqz');
for k = 1:numel(omega_list)
    fprintf('%18.6f %18.6f %18.6f\n', omega_list(k), w0eq_list(k), wpk_list(k));
end
```

digfil.dvi

(a)

$$y[n] + \frac{1}{6}y[n-1] - \frac{5}{48}y[n-2] - \frac{1}{48}y[n-3] = x[n] + x[n-1] - 6x[n-2].$$

Taking the z-transform using $\mathcal{Z}\{x[n-k]\} = z^{-k}X(z)$ and $\mathcal{Z}\{y[n-k]\} = z^{-k}Y(z)$:

$$Y(z)\Big(1+\frac{1}{6}z^{-1}-\frac{5}{48}z^{-2}-\frac{1}{48}z^{-3}\Big) = X(z)\Big(1+z^{-1}-6z^{-2}\Big).$$

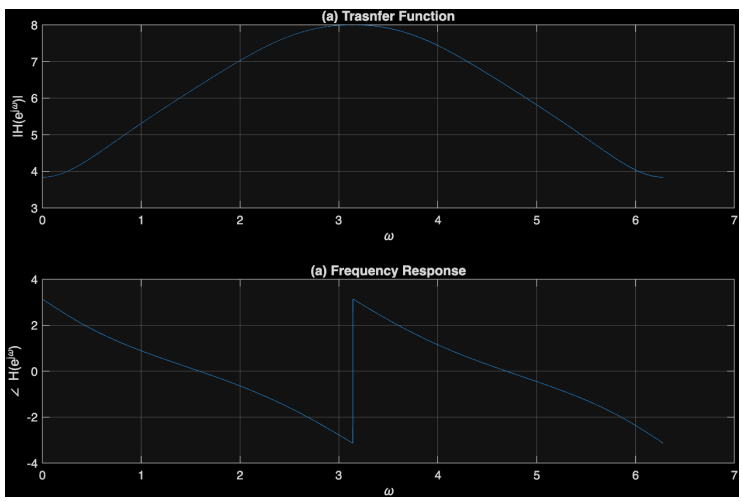
$$\boxed{H(z) \triangleq \frac{Y(z)}{X(z)} = \frac{1+z^{-1}-6z^{-2}}{1+\frac{1}{6}z^{-1}-\frac{5}{48}z^{-2}-\frac{1}{48}z^{-3}}}$$

Or also

$$H(z) = \frac{z^3+z^2-6z}{z^3+\frac{1}{6}z^2-\frac{5}{48}z-\frac{1}{48}}.$$

I got the frequency response by evaluating on the unit circle $z = e^{j\omega}$:

$$\boxed{H\big(e^{j\omega}\big) = \frac{1+e^{-j\omega}-6e^{-j2\omega}}{1+\frac{1}{6}e^{-j\omega}-\frac{5}{48}e^{-j2\omega}-\frac{1}{48}e^{-j3\omega}}}$$



(b)

From part (a),

$$H(z) = \frac{1+z^{-1}-6z^{-2}}{1+\frac{1}{6}z^{-1}-\frac{5}{48}z^{-2}-\frac{1}{48}z^{-3}} = \frac{z^2+z-6}{z^2}\cdot\frac{z^3}{z^3+\frac{1}{6}z^2-\frac{5}{48}z-\frac{1}{48}}.$$

Thus the zeros are the roots of

$$z^2+z-6=0 \quad\Rightarrow\quad (z-2)(z+3)=0$$

$$\boxed{z_1 = 2, \qquad z_2 = -3}$$

The poles are the roots of the denominator polynomial

$$z^3 + \frac{1}{6}z^2 - \frac{5}{48}z - \frac{1}{48} = 0.$$

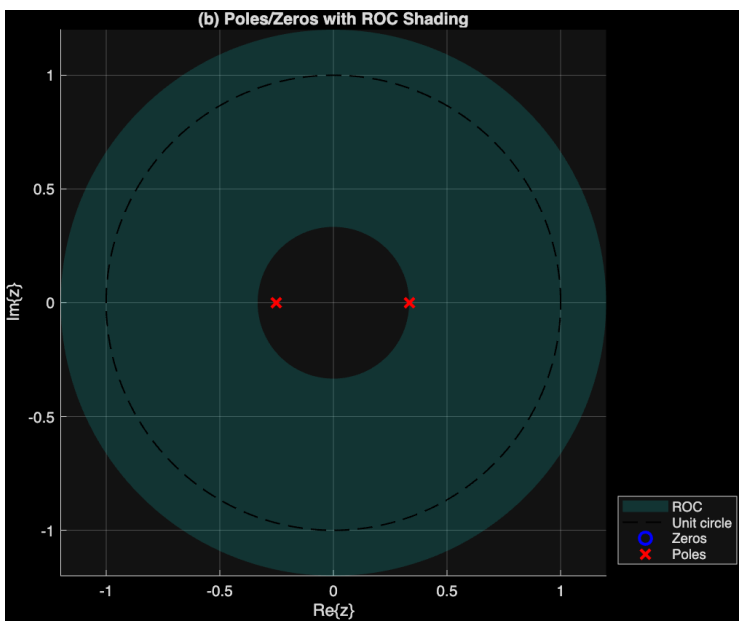
Factoring,

$$z^3 + \frac{1}{6}z^2 - \frac{5}{48}z - \frac{1}{48} = \left(z - \frac{1}{3}\right)\left(z + \frac{1}{4}\right)^2,$$

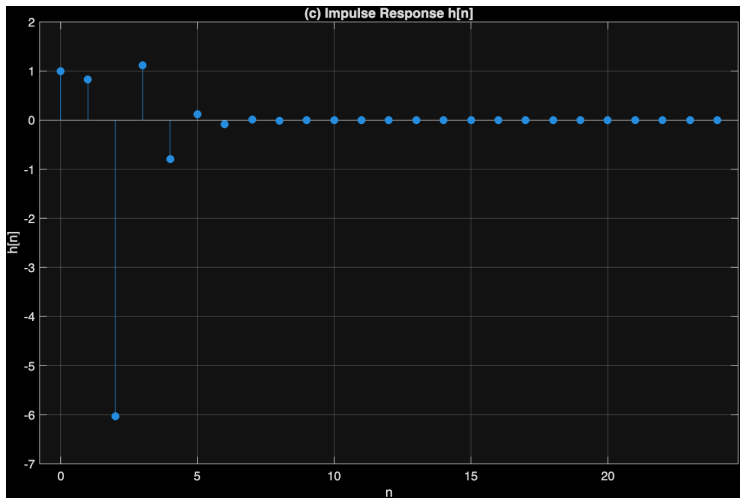
$$\boxed{p_1 = \frac{1}{3}, \qquad p_2 = p_3 = -\frac{1}{4} \ \text{(double pole)}}$$

Since the system is causal, the ROC is outside the outermost pole:

$$\boxed{\text{ROC:} \quad |z| > \max\{|p_k|\} = \max\left\{\frac{1}{3}, \frac{1}{4}\right\} = \frac{1}{3}}$$



(c)



(d)

I applied the delayed step input $x[n] = u[n-5]$ to the system and calculated the output $y[n]$ in MATLAB using the difference equation (`filter(b,a,x)`), and then plotted $y[n]$ versus n. I found that after the step occurred at $n = 5$, the output transient decayed and $y[n]$ approached a constant steady-state value.

I found the analytical DC gain as

$$H(1) = \left.\frac{1 + z^{-1} - 6z^{-2}}{1 + \frac{1}{6}z^{-1} - \frac{5}{48}z^{-2} - \frac{1}{48}z^{-3}}\right|_{z=1} = \frac{1+1-6}{1+\frac{1}{6}-\frac{5}{48}-\frac{1}{48}} = \frac{-4}{\frac{25}{24}} = -\frac{96}{25} = -3.84.$$

Since $x[n] = u[n-5]$ has unit DC level, I expected the steady-state output to be $y[n] \to H(1) \cdot 1 = -3.84$.

$$y_{\text{ss}} \approx -3.83998586165.$$

I compared this to the analytical DC gain and found the difference

$$y_{\text{ss}} - H(1) \approx 1.41383539876 \times 10^{-5},$$

thus the observed DC gain agreed with the analytical DC gain.

(e)

$$x[n] = \big(4 + \cos(0.25\pi n)\big)u[n] = \big(4 + \cos(\tfrac{\pi}{4}n)\big)u[n].$$

I first rewrote the cosine using complex exponentials:

$$\cos\left(\tfrac{\pi}{4}n\right) = \frac{1}{2}\left(e^{j\frac{\pi}{4}n} + e^{-j\frac{\pi}{4}n}\right).$$

$$x[n] = 4u[n] + \frac{1}{2}e^{j\frac{\pi}{4}n}u[n] + \frac{1}{2}e^{-j\frac{\pi}{4}n}u[n].$$

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}.$$

Since $u[n] = 0$ for $n < 0$, I reduced the sum to $n \geq 0$:

$$X(e^{j\omega}) = \sum_{n=0}^{\infty}\left(4 + \cos\left(\tfrac{\pi}{4}n\right)\right)e^{-j\omega n}.$$

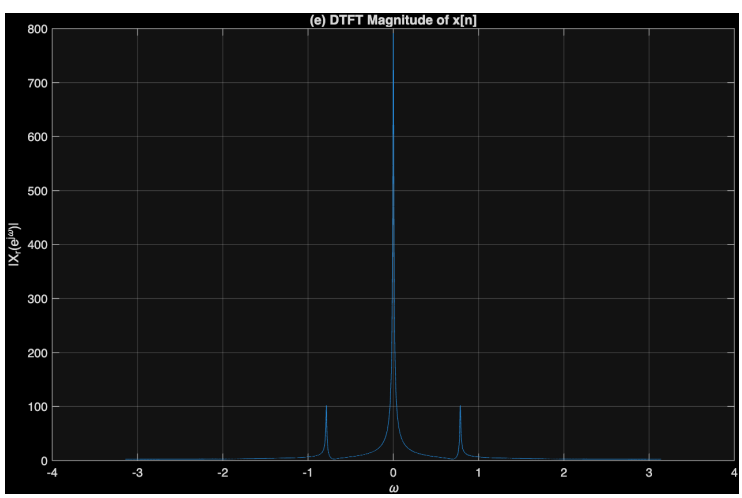
Using the exponential form and linearity,

$$\begin{aligned} X(e^{j\omega}) &= 4\sum_{n=0}^{\infty} e^{-j\omega n} + \frac{1}{2}\sum_{n=0}^{\infty} e^{j\frac{\pi}{4}n}e^{-j\omega n} + \frac{1}{2}\sum_{n=0}^{\infty} e^{-j\frac{\pi}{4}n}e^{-j\omega n} \\ &= 4\sum_{n=0}^{\infty}\left(e^{-j\omega}\right)^n + \frac{1}{2}\sum_{n=0}^{\infty}\left(e^{-j(\omega-\frac{\pi}{4})}\right)^n + \frac{1}{2}\sum_{n=0}^{\infty}\left(e^{-j(\omega+\frac{\pi}{4})}\right)^n. \end{aligned}$$

Each term is a geometric series. I used $z = re^{j\omega}$ with $0 < r < 1$:

$$\sum_{n=0}^{\infty}\left(re^{-j\theta}\right)^n = \frac{1}{1 - re^{-j\theta}}, \qquad 0 < r < 1.$$

$$\boxed{X_r(e^{j\omega}) = \frac{4}{1 - re^{-j\omega}} + \frac{1}{2}\frac{1}{1 - re^{-j(\omega-\frac{\pi}{4})}} + \frac{1}{2}\frac{1}{1 - re^{-j(\omega+\frac{\pi}{4})}}, \qquad 0 < r < 1}$$



(f)

$$Y(e^{j\omega}) = H(e^{j\omega})\, X(e^{j\omega}).$$

From part (a), I used

$$H(e^{j\omega}) = \frac{1 + e^{-j\omega} - 6e^{-j2\omega}}{1 + \frac{1}{6}e^{-j\omega} - \frac{5}{48}e^{-j2\omega} - \frac{1}{48}e^{-j3\omega}}.$$

From part (e), for $x[n] = (4 + \cos(\frac{\pi}{4}n))u[n]$, I wrote the (regularized) DTFT as

$$X_r(e^{j\omega}) = \frac{4}{1 - re^{-j\omega}} + \frac{1}{2}\frac{1}{1 - re^{-j(\omega - \frac{\pi}{4})}} + \frac{1}{2}\frac{1}{1 - re^{-j(\omega + \frac{\pi}{4})}}, \qquad 0 < r < 1.$$

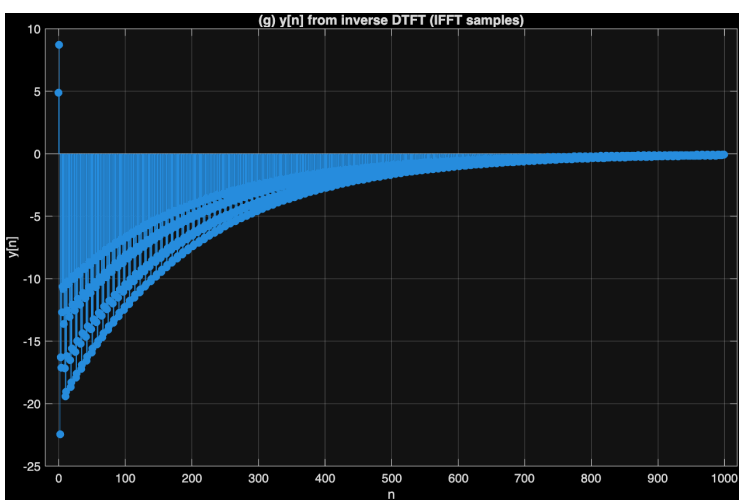
Therefore, by direct multiplication,

$$\begin{aligned} Y_r(e^{j\omega}) &= H(e^{j\omega})\, X_r(e^{j\omega}) \\ &= \frac{1 + e^{-j\omega} - 6e^{-j2\omega}}{1 + \frac{1}{6}e^{-j\omega} - \frac{5}{48}e^{-j2\omega} - \frac{1}{48}e^{-j3\omega}} \\ &\quad \times \left[\frac{4}{1 - re^{-j\omega}} + \frac{1}{2}\frac{1}{1 - re^{-j(\omega - \frac{\pi}{4})}} + \frac{1}{2}\frac{1}{1 - re^{-j(\omega + \frac{\pi}{4})}}\right], \qquad 0 < r < 1. \end{aligned}$$

(g)

I first sampled $Y(e^{j\omega}) = H(e^{j\omega})X(e^{j\omega})$ uniformly on a discrete frequency grid $\omega_k = 2\pi k/N_{\text{fft}}$. I then used the inverse FFT to approximate the inverse DTFT and obtained the time-domain output sequence $y[n]$.

$$y_{\text{IFFT}}[n] = r^n \left(-15.36 + \left|H\left(e^{j\pi/4}\right)\right| \cos\left(\frac{\pi}{4}n + \angle H\left(e^{j\pi/4}\right)\right)\right) + \text{transients}.$$



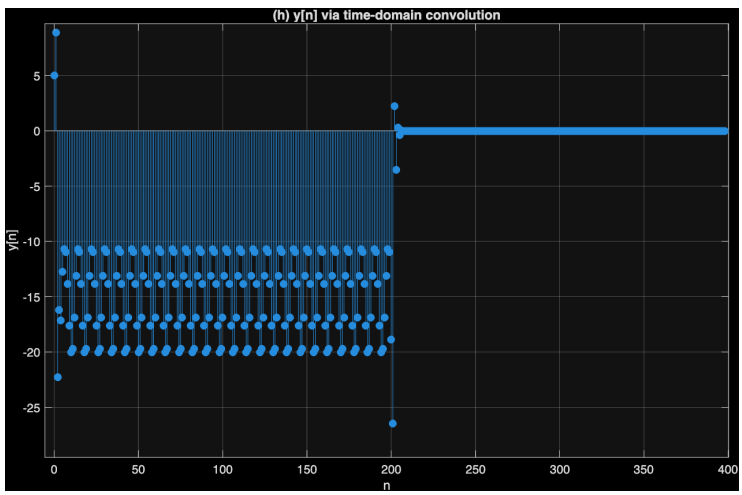
(h)

The maximum absolute difference between the two outputs over the first 399 samples was

$$\max_n \left| y_{\text{conv}}[n] - y_{\text{IDTFT}}[n] \right| = 1.07 \times 10^{-14},$$

which is on the order of numerical round-off error.

Therefore, the time-domain convolution result agrees with the inverse-DTFT result from part (g).



(i)

$$H(e^{j\omega}) = \frac{1 + e^{-j\omega} - 6e^{-j2\omega}}{1 + \frac{1}{6}e^{-j\omega} - \frac{5}{48}e^{-j2\omega} - \frac{1}{48}e^{-j3\omega}}.$$

First, at $\omega = 0$,

$$H(e^{j0}) = \frac{1 + 1 - 6}{1 + \frac{1}{6} - \frac{5}{48} - \frac{1}{48}} = \frac{-4}{\frac{25}{24}} = -\frac{96}{25} = -3.84.$$

Next, at $\omega = \frac{\pi}{4}$,

$$H(e^{j\pi/4}) = \frac{1 + e^{-j\pi/4} - 6e^{-j\pi/2}}{1 + \frac{1}{6}e^{-j\pi/4} - \frac{5}{48}e^{-j\pi/2} - \frac{1}{48}e^{-j3\pi/4}}.$$

This yieled a complex number:

$$H(e^{j\pi/4}) = \left| H(e^{j\pi/4}) \right| e^{j\angle H(e^{j\pi/4})}.$$

Since the input signal is

$$x[n] = \left(4 + \cos(\tfrac{\pi}{4}n)\right)u[n],$$

it contains a DC component and a sinusoidal component at frequency $\omega = \frac{\pi}{4}$.

I know that the DC component of the output was scaled by the DC gain $H(e^{j0})$, while the sinusoidal component retained its frequency and was scaled in magnitude by $\left|H(e^{j\pi/4})\right|$ and phase-shifted by $\angle H(e^{j\pi/4})$.

Thus, the steady-state output can be written as

$$y_{\text{ss}}[n] = 4H(e^{j0}) + \left|H(e^{j\pi/4})\right| \cos\left(\frac{\pi}{4}n + \angle H(e^{j\pi/4})\right).$$

I used the fact that complex exponentials are eigenfunctions of LTI systems to analytically determine the steady-state response of the system.

code for part2

```
%% part (a)
b = [1 1 -6];
a = [1 1/6 -5/48 -1/48];

disp('H(z) = (1 + z^{-1} - 6z^{-2}) / (1 + (1/6)z^{-1} - (5/48)z^{-2} - (1/48)z^{-3})')

[H,w] = freqz(b,a,4096,'whole');

figure
subplot(2,1,1)
plot(w,abs(H)), grid on
xlabel('\omega'), ylabel('|H(e^{j\omega})|')
title('(a) Trasnfer Function')


subplot(2,1,2)
plot(w,angle(H)), grid on
xlabel('\omega'), ylabel('\angle H(e^{j\omega})')
title('(a) Frequency Response')


%% part (b)
z0 = roots(b);
p0 = roots(a);

disp('Zeros (roots of numerator):')
disp(z0)
disp('Poles (roots of denominator):')
disp(p0)
```

```
rmax = max(abs(p0));
fprintf('Causal ROC: |z| > %.12g\n', rmax)

th = linspace(0,2*pi,1200);
Rout = max(1.2, 1.5*rmax);

xout = Rout*cos(th); yout = Rout*sin(th);
xin  = rmax*cos(th); yin  = rmax*sin(th);

figure
patch([xout fliplr(xin)], [yout fliplr(yin)], 1, 'FaceAlpha', 0.2, 'EdgeColor', 'none')
hold on
plot(cos(th), sin(th), 'k--', 'LineWidth', 1)
plot(real(z0), imag(z0), 'bo', 'LineWidth', 2, 'MarkerSize', 8)
plot(real(p0), imag(p0), 'rx', 'LineWidth', 2, 'MarkerSize', 9)
grid on
axis equal
axis([-Rout Rout -Rout Rout])
xlabel('Re\{z\}')
ylabel('Im\{z\}')
title('(b) Poles/Zeros with ROC Shading')
legend('ROC','Unit circle','Zeros','Poles','Location','best')

%% part (c)
N = 25;
[h,n] = impz(b,a,N);

figure
stem(n,h,'filled')
grid on
xlabel('n')
ylabel('h[n]')
title('(c) Impulse Response h[n]')

%% part (d)
N = 25;
n = 0:N-1;
x = double(n >= 5);

y = filter(b,a,x);

figure
stem(n,y,'filled')
grid on
xlabel('n')
ylabel('y[n]')
```

```
title('(d) Output y[n] for x[n] = u[n-5]')

Hdc = sum(b)/sum(a);
fprintf('Analytical DC gain H(1) = %.12g\n', Hdc)

yss = mean(y(end-9:end));
fprintf('Estimated steady-state y[n] (avg last 10 samples) = %.12g\n', yss)
fprintf('Difference (y_ss - H(1)) = %.12g\n', yss - Hdc)

%% part (e)
w0 = pi/4;
r  = 0.995;
Nw = 4096;
w  = linspace(-pi,pi,Nw);

% I wanted to show the evaluation of the DTFT too.

fprintf('DTFT derivation for x[n] = (4 + cos(0.25*pi*n))u[n]\n\n');
fprintf('x[n] = 4u[n] + (1/2)e^{j(/4)n}u[n] + (1/2)e^{-j(/4)n}u[n]\n\n');

fprintf('DTFT{u[n]} = 1 / (1 - r e^{-j}),   |r| < 1\n\n');

fprintf('X_r(e^{j}) = 4 / (1 - r e^{-j})\n');
fprintf('           + (1/2) / (1 - r e^{-j( - /4)})\n');
fprintf('           + (1/2) / (1 - r e^{-j( + /4)})\n\n');

fprintf('Regularization radius r = %.3f\n\n', r);

X = 4./(1 - r*exp(-1j*w)) ...
  + 0.5./(1 - r*exp(-1j*(w - w0))) ...
  + 0.5./(1 - r*exp(-1j*(w + w0)));

figure
plot(w,abs(X)), grid on
xlabel('\omega')
ylabel('|X_r(e^{j\omega})|')
title('(e) DTFT Magnitude of x[n]')

%% part (f) + (g)
Nfft = 1000;
k = 0:Nfft-1;
w = 2*pi*k/Nfft;

w0 = pi/4;
r  = 0.995;
```

```
H = (b(1) + b(2)*exp(-1j*w) + b(3)*exp(-1j*2*w)) ./ ...
    (a(1) + a(2)*exp(-1j*w) + a(3)*exp(-1j*2*w) + a(4)*exp(-1j*3*w));

X = 4./(1 - r*exp(-1j*w)) ...
  + 0.5./(1 - r*exp(-1j*(w - w0))) ...
  + 0.5./(1 - r*exp(-1j*(w + w0)));

Y = H .* X;

y_ifft = real(ifft(Y));
n_ifft = 0:Nfft-1;

figure
stem(n_ifft, y_ifft, 'filled')
grid on
xlabel('n')
ylabel('y[n]')
title('(g) y[n] from inverse DTFT (IFFT samples)')


%% part (h)
Nx = 200;
n = 0:Nx-1;
x = (4 + cos((pi/4)*n)) .* (n>=0);
x = x(:);

Nh = 200;
h = impz(b,a,Nh);
h = h(:);

y_conv = conv(x,h);
L = length(y_conv);

Nfft = 2^nextpow2(L);
y_fft_full = ifft( fft(x,Nfft) .* fft(h,Nfft) );
y_fft = real(y_fft_full(1:L));

err = y_conv - y_fft;
fprintf('(h) Max abs difference (conv vs FFT) over first %d samples: \\
%.12g\n', L, max(abs(err)));

figure
stem(0:L-1, y_conv, 'filled')
grid on
xlabel('n')
ylabel('y[n]')
```

```
title('(h) y[n] via time-domain convolution')
```